Rational Functions and Asymptotes

ANSWER KEY

Section 1: Vertical Asymptotes and Holes

1 Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 4}{x^2 - x - 2}$.

a. **Hole at $x = 2$, Vertical asymptote at $x = -1$** ✓

b. Vertical asymptotes at $x = 2$ and $x = -1$

c. Hole at $x = -1$, Vertical asymptote at $x = 2$

d. No holes, vertical asymptote at $x = 2$

Factor: $\frac{(x-2)(x+2)}{(x-2)(x+1)} = \frac{x+2}{x+1}$ for $x \neq 2$. The common factor $(x - 2)$ creates a hole at $x = 2$. The remaining zero of the denominator $x = -1$ gives a vertical asymptote.

2 Find the coordinates (x, y) of the hole for $f(x) = \frac{x^2 - 9}{x - 3}$.

a. $(3, 6)$ ✓

b. $(3, 0)$

c. $(3, 3)$

d. $(-3, 0)$

For $x \neq 3$, $f(x) = x + 3$. Evaluating at $x = 3$ gives $y = 3 + 3 = 6$. The hole is at $(3, 6)$.

3 How do you determine if a real zero of the denominator causes a hole or a vertical asymptote?

a. **If the numerator also has the same zero of equal or greater multiplicity, it is a hole; otherwise it is a vertical asymptote** ✓

b. Every zero of the denominator is an asymptote

c. If the degree of the numerator is larger, it is always a hole

d. Check the sign of $f(0)$

A factor that cancels completely between numerator and denominator creates a removable hole; an uncanceled denominator factor yields an infinite discontinuity (vertical asymptote).

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Section 2: Horizontal Asymptotes

4 Find the horizontal asymptote of $f(x) = \frac{4x^2 + 1}{3x^2 - 5x + 2}$.

a. $y = 4/3$ ✓

b. $y = 0$

c. No horizontal asymptote

d. $y = 1/2$

The degrees of numerator and denominator are both 2. The horizontal asymptote is the ratio of leading coefficients: $y = 4/3$.

5 Find the horizontal asymptote of $g(x) = \frac{7x - 2}{x^3 + 4}$.

a. $y = 0$ ✓

b. $y = 7$

c. No horizontal asymptote

d. $y = -1/2$

The degree of the numerator (1) is strictly less than the degree of the denominator (3), so $y = 0$ is the horizontal asymptote.

6 Under what condition does a rational function have NO horizontal asymptote?

a. **When the degree of the numerator is strictly greater than the degree of the denominator** ✓

b. When the degree of the denominator is greater

c. When the degrees are equal

d. When there are no real roots

If $\deg(P) > \deg(Q)$, $f(x) \to \pm\infty$ as $x \to \pm\infty$, so there is no horizontal asymptote.

Section 3: Slant (Oblique) Asymptotes

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7 When does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. **When the degree of the numerator is exactly 1 greater than the degree of the denominator ✓**

b. When the degree of the numerator is 2 greater

c. When the degree of the denominator is greater

d. When there are no vertical asymptotes

A slant asymptote occurs when $\deg(P) = \deg(Q) + 1$, producing a linear quotient via polynomial division.

8 Find the slant asymptote of $f(x) = \frac{2x^2 + 7x + 5}{x + 2}$.

a. $y = 2x + 3$ ✓

b. $y = 2x + 7$

c. $y = 2x - 1$

d. $y = x + 3$

Polynomial division: $\frac{2x^2 + 7x + 5}{x + 2} = (2x + 3) - \frac{1}{x + 2}$. As $x \to \pm\infty$, the remainder $\to 0$, giving slant asymptote $y = 2x + 3$.

9 Find the slant asymptote of $g(x) = \frac{x^3 - 1}{x^2 + 1}$.

a. $y = x$ ✓

b. $y = x - 1$

c. $y = 1$

d. No slant asymptote

Dividing: $x^3 - 1 = x(x^2 + 1) - x - 1 \Rightarrow \frac{x^3 - 1}{x^2 + 1} = x - \frac{x + 1}{x^2 + 1}$. As $x \to \infty$, slant asymptote is $y = x$.

Section 4: Intercepts and Asymptote Crossings

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10 Find all x-intercepts and the y-intercept of $f(x) = \frac{x^2 - 1}{x^2 - 4}$.

a. **x-intercepts:** $(\pm 1, 0)$**, y-intercept:** $(0, 1/4)$ ✓

b. x-intercepts: $(\pm 2, 0)$, y-intercept: $(0, -1/4)$

c. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, -1)$

d. x-intercepts: $(1, 0)$, y-intercept: $(0, 1/4)$

x-intercepts occur where numerator $= 0$: $x^2 - 1 = 0 \Rightarrow x = \pm 1$. y-intercept is $f(0) = \frac{-1}{-4} = 1/4$.

11 Can a rational function cross its horizontal or slant asymptote?

a. **Yes, a function can cross its horizontal or slant asymptote at finite values of** x ✓

b. No, a graph can never touch any asymptote

c. Only at the y-intercept

d. Only if it has no vertical asymptotes

Unlike vertical asymptotes (where the function is undefined), horizontal and slant asymptotes only govern end behavior as $x \to \pm\infty$; curves can freely cross them at finite points.

12 Find the point where $f(x) = \frac{x^2 + 3x - 4}{x^2 - 1}$ crosses its horizontal asymptote $y = 1$.

a. $(-1, 1)$ is undefined; crossing is at $x = 1$ is undefined; crosses at $(1, 1)$ is a hole; no crossing point

b. It crosses at $x = 1$

c. It crosses at $x = 0$

d. **It does not cross its horizontal asymptote** ✓

Set $f(x) = 1$: $\frac{x^2 + 3x - 4}{x^2 - 1} = 1 \Rightarrow x^2 + 3x - 4 = x^2 - 1 \Rightarrow 3x = 3 \Rightarrow x = 1$. But at $x = 1$, the function has a removable hole, so it does not cross.

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